CAISO API Program Concept

HAILEY HENDRICKSON

DATA INTERN

SUMMER 2025 (MAY 12- AUG 14)

GOALS

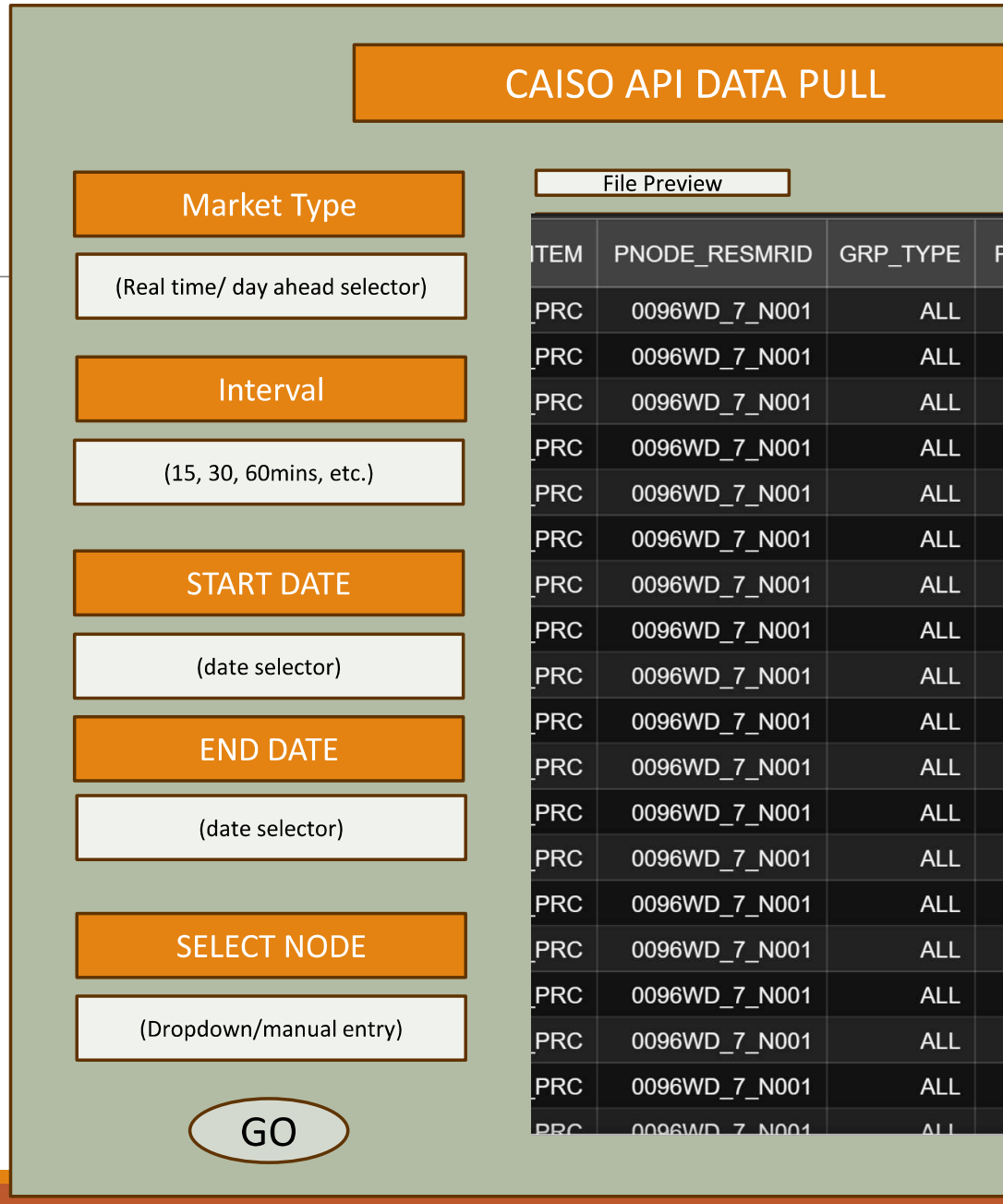
GOAL: Create a GUI that simplifies bulk data pulling for consultants and analysts withou
knowledge

Sub-goals:

- Increase understanding of CAISO data/ energy market as a whole
- Perform simple analysis of data pulls

FUNCTION

1. Consultants/Analysts will access the GUI and input with their desired pull settings
2. Code function:
   - pulls data using the CAISO API
   - combines separate API pulls (can only do 30 days at a time)
   - Fixes column names for clarity
   - Fill in missing hours using previous day's value
3. Outputs: a preview of the file and a link to download the CSV



RESOURCES

Software platform

- Coding: VSC
- GUI: either the free python GUI (Tkinter) or Streamlit (web hosted?)
  - Depends on needs and ease of use. Web could be easier for access then sending python code to be ran

Tools/ Resources

- CAISO API
- Mentors
  - Zoe, Malkie

WORKFLOW/STEPS

1- finish project concept ✅

2- baseline CAISO API query in python and data dictionary ✅

3- build cleaning program ✅

4- finish bulk of cleaning, create “minimum viable product”: push code to GUI of choice ✅

5-finish “minimum viable product” ✅

6- add to product, get feedback from consultants on usability ✅

7- finish full product deliverable, get more feedback ✅

8- revise final deliverable ✅

9- create supporting documentation and walk-through video, finish HMA report ✅

PROCESS

In the next few slides you will see snippets of the work I have done, all leading up to th
product.

Learning the Energy Market: What is CAISO

CAISO stands for the California Independent System O

They are an organization that helps to manage the flo across California. They also publish energy data, inclu price info, transmission info, supply and demand, and information to help generators, utilities, and other sta make informed decisions.

Locational Marginal Pricing

Takes into account 3 components:

- cost of energy (generation)

- cost of loss (during transmission)

- cost of congestion (when there isn't enough infrastructure to transfer energy from a g
to a consumer)

Likely for more congestion costs as renewables take over more (hard to transfer these
better infrastructure)

FUNCTIONS

imports

Calling the API

Printing the individual files

Cleaning the files (separately)

Map to find the right report type

User inputs

Breaking the date range into chunks and calling my functions

Combining cleaned files to one final product

Imports

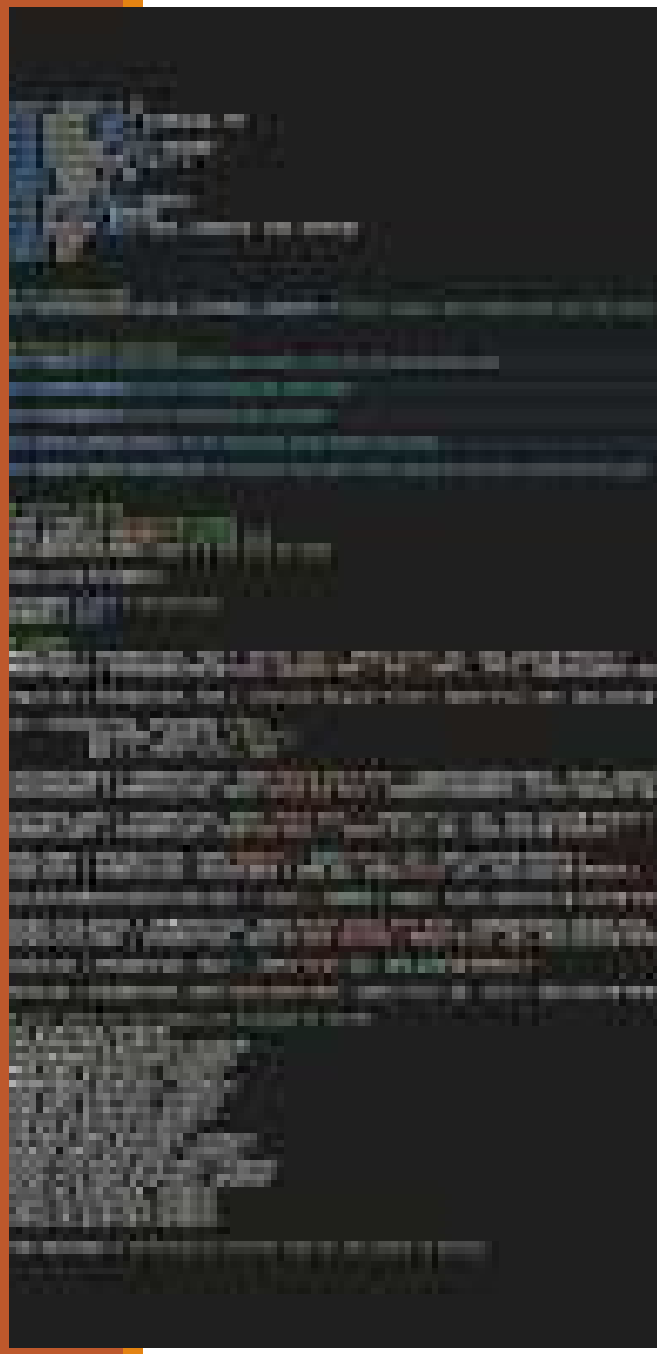
Functions
- Calls backend code
- Widget Button Functions

Opening GUI

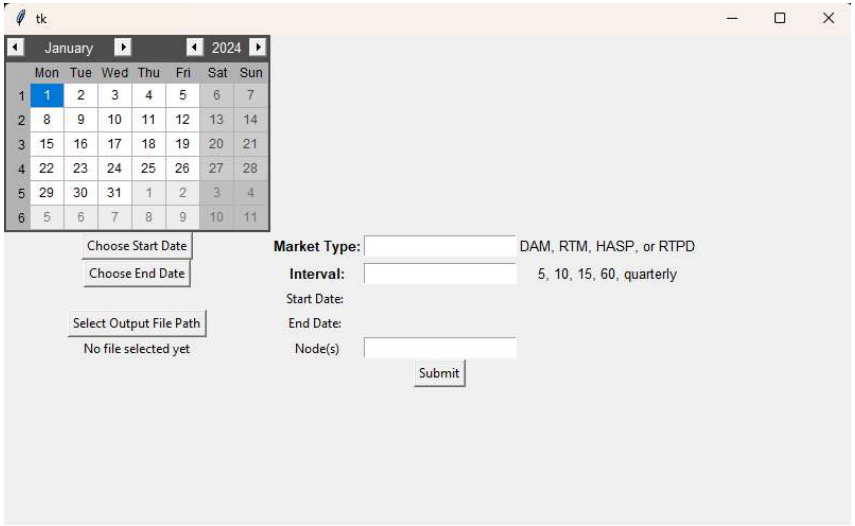
Widgets for GUI

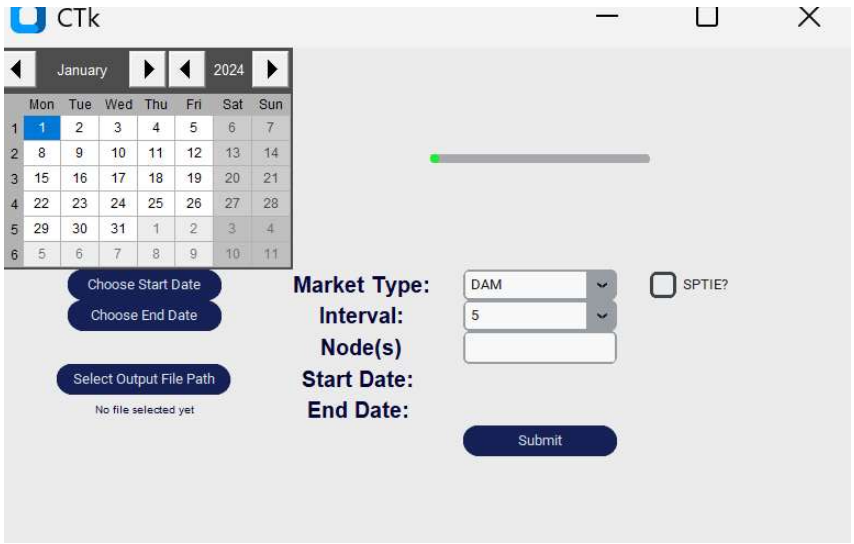
Creating Layout

Line that runs it all



* 667 lines of code in total!



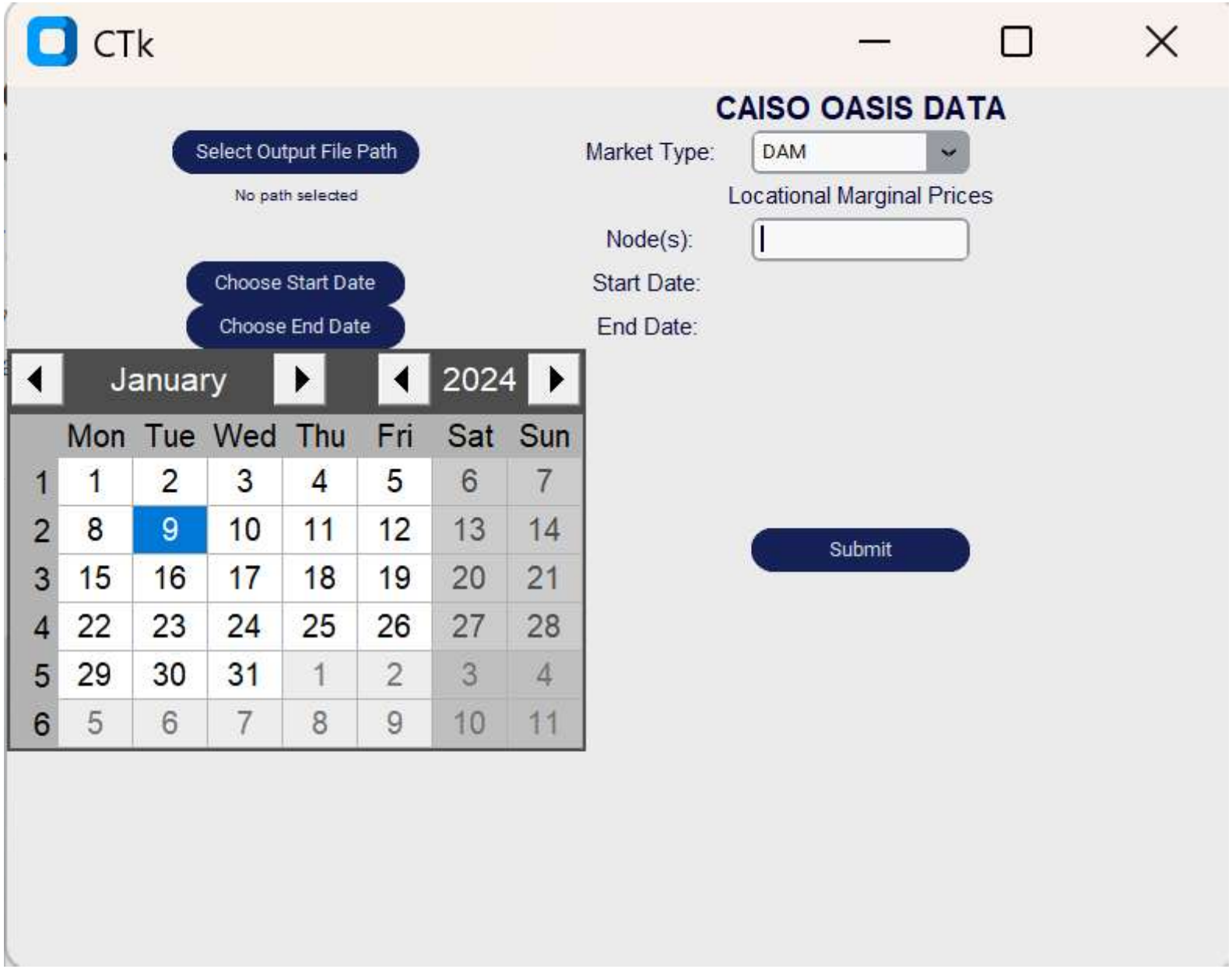


GUI Drafts

This is a screenshot of the first draft of my GUI
successfully pulls and cleans each report type.

It doesn't look pretty, but it is functional!

The second draft uses a slightly different versio
tkinter, the python graphic user interface creat
prettier, same functionality.



Final GUI

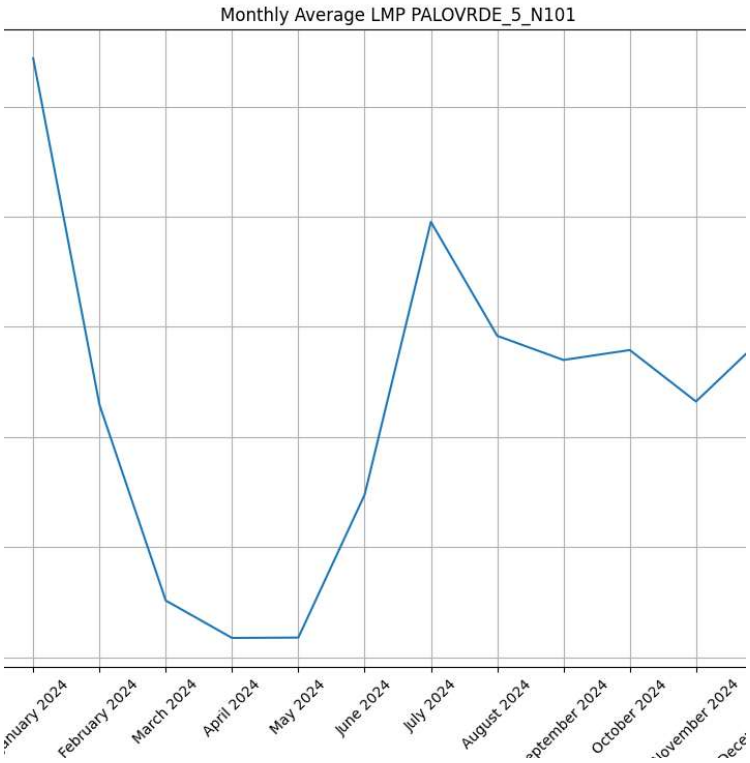
Reorganized widgets
status label at the bo
Functionality is impr

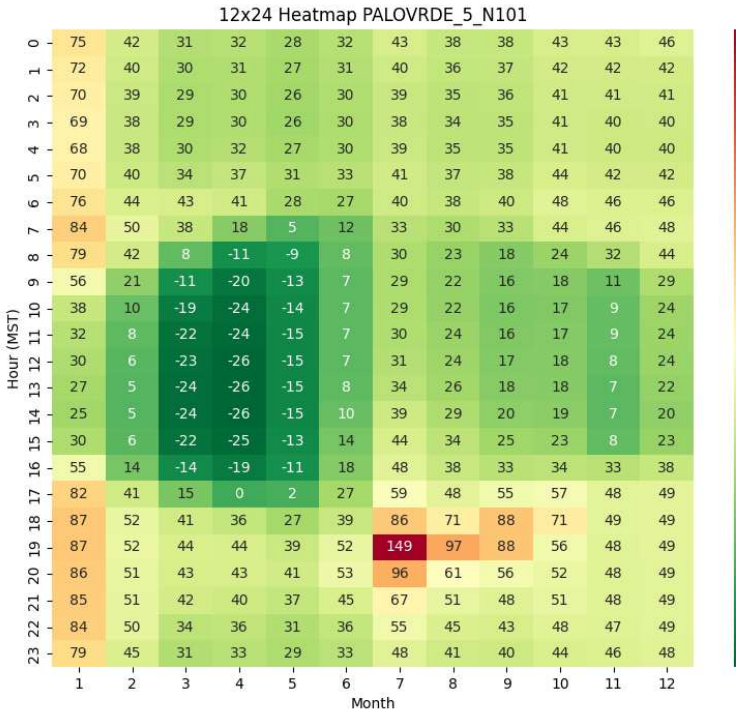
1	INTERVALSTARTTIME_GMT	INTERVALENDTIME_GMT	NODE	Year	Month	Day	Hour (GMT)	Minute	Congestion	Energy	eenhouse G	Loss	LM
238	2024-01-13 21:00:00	2024-01-13 21:15:00	PALOVRDE_	2024	1	13	21	0	-406.493	442.8474	-19.29	-10.9278	6.
239	2024-01-13 21:15:00	2024-01-13 21:30:00	PALOVRDE_	2024	1	13	21	15	-523.933	586.7567	-40.88	-12.0639	9.
240	2024-01-13 21:30:00	2024-01-13 21:45:00	PALOVRDE_	2024	1	13	21	30	-469.439	522.1451	-40.07	-11.3288	1.
241	2024-01-13 21:45:00	2024-01-13 22:00:00	PALOVRDE_	2024	1	13	21	45	-494.185	533.2239	-23.9019	-11.7653	3.
242	2024-01-13 22:00:00	2024-01-13 22:15:00	PALOVRDE_	2024	1	13	22	0	-473.612	516.7512	-35.876	-11.2524	-3.
243	2024-01-13 22:15:00	2024-01-13 22:30:00	PALOVRDE_	2024	1	13	22	15	-234.594	250.7553	-18.84	-6.1689	-8.
244	2024-01-13 22:30:00	2024-01-13 22:45:00	PALOVRDE_	2024	1	13	22	30	-403.453	449.4012	-19.29	-11.699	14.
245	2024-01-13 22:45:00	2024-01-13 23:00:00	PALOVRDE_	2024	1	13	22	45	-469.654	539.1473	-19	-13.9919	36.
261	2024-01-14 02:45:00	2024-01-14 03:00:00	PALOVRDE_	2024	1	14	2	45	-468.404	656.2261	-40.88	-9.8849	137.
262	2024-01-14 03:00:00	2024-01-14 03:15:00	PALOVRDE_	2024	1	14	3	0	-529.876	737.448	-41.64	-10.7186	155.
263	2024-01-14 03:15:00	2024-01-14 03:30:00	PALOVRDE_	2024	1	14	3	15	-561.148	761.9218	-40.07	-11.2264	149.
264	2024-01-14 03:30:00	2024-01-14 03:45:00	PALOVRDE_	2024	1	14	3	30	-581.698	770.2696	-40.07	-11.0534	137.
265	2024-01-14 03:45:00	2024-01-14 04:00:00	PALOVRDE_	2024	1	14	3	45	-570.327	768.0429	-40.07	-11.6395	146.
266	2024-01-14 04:00:00	2024-01-14 04:15:00	PALOVRDE_	2024	1	14	4	0	-384.046	643.7666	-41.64	-9.6942	208.
267	2024-01-14 04:15:00	2024-01-14 04:30:00	PALOVRDE_	2024	1	14	4	15	-555.644	741.8987	-40.88	-9.6036	135.
268	2024-01-14 04:30:00	2024-01-14 04:45:00	PALOVRDE_	2024	1	14	4	30	-446.79	616.4059	-40.07	-7.5052	122.
269	2024-01-14 04:45:00	2024-01-14 05:00:00	PALOVRDE	2024	1	14	4	45	-393.149	549.4133	-40.07	0	116.

RTPD | Hourly Average

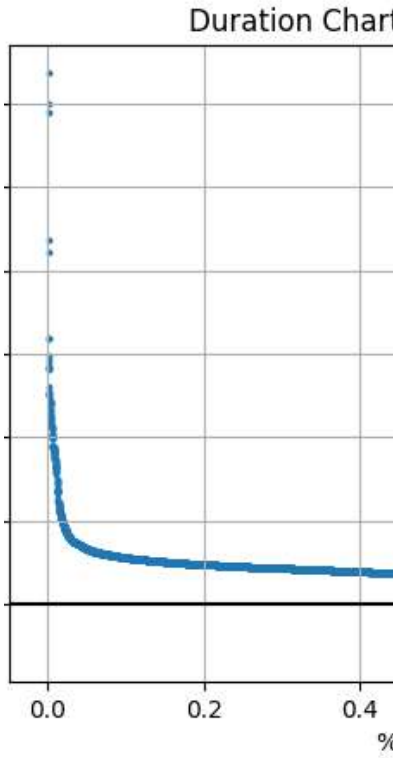
OUTPUT

HAS A PAGE FOR THE REPORT DATA, AS WELL AS A FEW ANALYSIS PAGES.



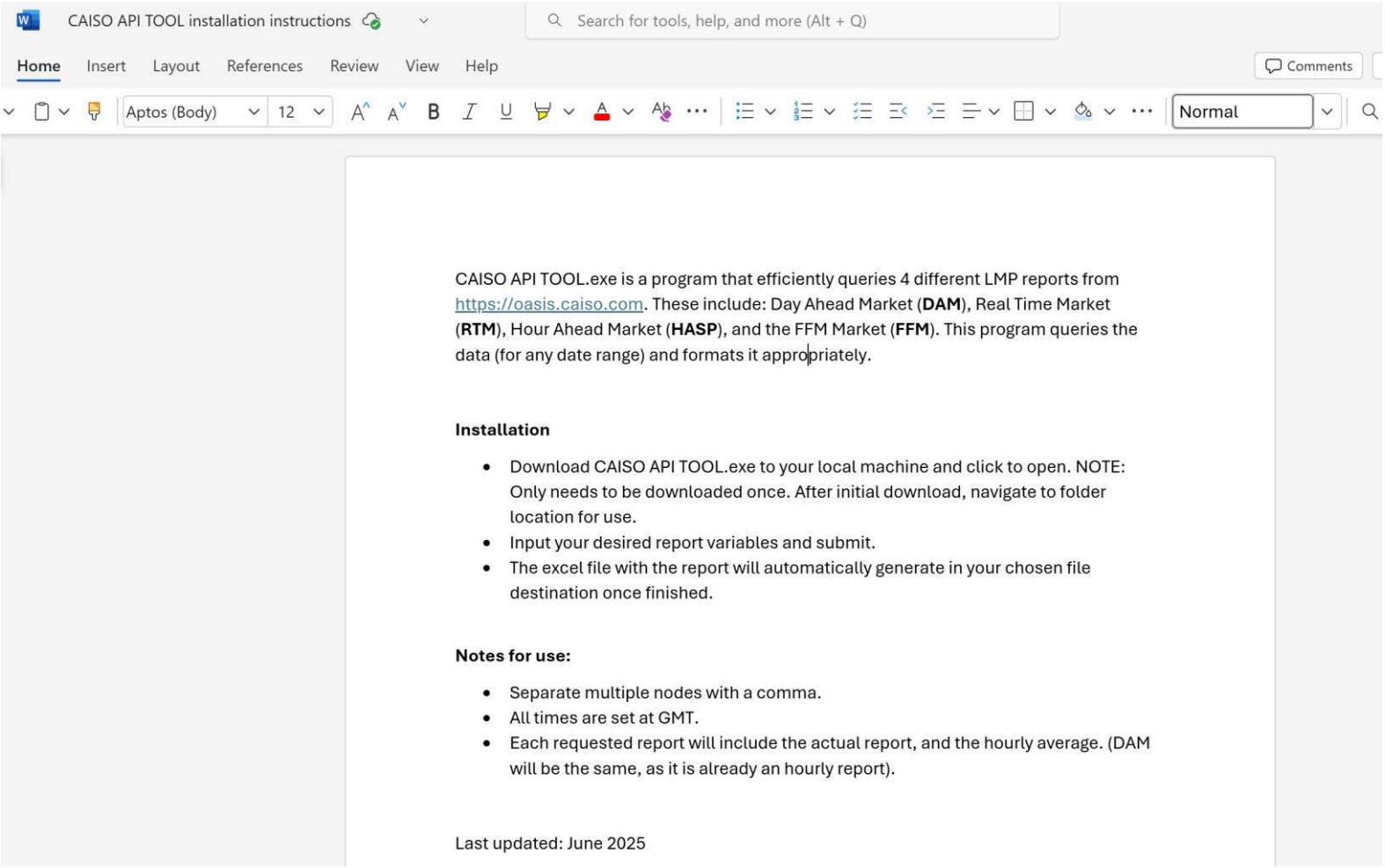


Hour (MST)	1	2	3	4	5	6	7	8	9	10	11	12
0	75	42	31	32	28	32	43	38	38	43	43	46
1	72	40	30	31	27	31	40	36	37	42	42	42
2	70	39	29	30	26	30	39	35	36	41	41	41
3	69	38	29	30	26	30	38	34	35	41	40	40
4	68	38	30	32	27	30	39	35	35	41	40	40
5	70	40	34	37	31	33	41	37	38	44	42	42
6	76	44	43	41	28	27	40	38	40	48	46	46
7	84	50	38	18	5	12	33	30	33	44	46	48
8	79	42	8	-11	-9	8	30	23	18	24	32	44
9	56	21	-11	-20	-13	7	29	22	16	18	11	29
10	38	10	-19	-24	-14	7	29	22	16	17	9	24
11	32	8	-22	-24	-15	7	30	24	16	17	9	24
12	30	6	-23	-26	-15	7	31	24	17	18	8	24
13	27	5	-24	-26	-15	8	34	26	18	18	7	22
14	25	5	-24	-26	-15	10	39	29	20	19	7	20
15	30	6	-22	-25	-13	14	44	34	25	23	8	23
16	55	14	-14	-19	-11	18	48	38	33	34	33	38
17	82	41	15	0	2	27	59	48	55	57	48	49
18	87	52	41	36	27	39	86	71	88	71	49	49
19	87	52	44	44	39	52	149	97	88	56	48	49
20	86	51	43	43	41	53	96	61	56	52	48	49
21	85	51	42	40	37	45	67	51	48	51	48	49
22	84	50	34	36	31	36	55	45	43	48	47	49
23	79	45	31	33	29	33	48	41	40	44	46	48



OUTPUT (ANALYSIS)

MONTHLY AVERAGE LINE CHART, 12X24 HOURLY HEAT MAP, DURATION CHARTS



CAISO API TOOL.exe is a program that efficiently queries 4 different LMP reports from https://oasis.caiso.com. These include: Day Ahead Market (**DAM**), Real Time Market (**RTM**), Hour Ahead Market (**HASP**), and the FFM Market (**FFM**). This program queries the data (for any date range) and formats it appropriately.

Installation

- Download CAISO API TOOL.exe to your local machine and click to open. NOTE: Only needs to be downloaded once. After initial download, navigate to folder location for use.
- Input your desired report variables and submit.
- The excel file with the report will automatically generate in your chosen file destination once finished.

Notes for use:

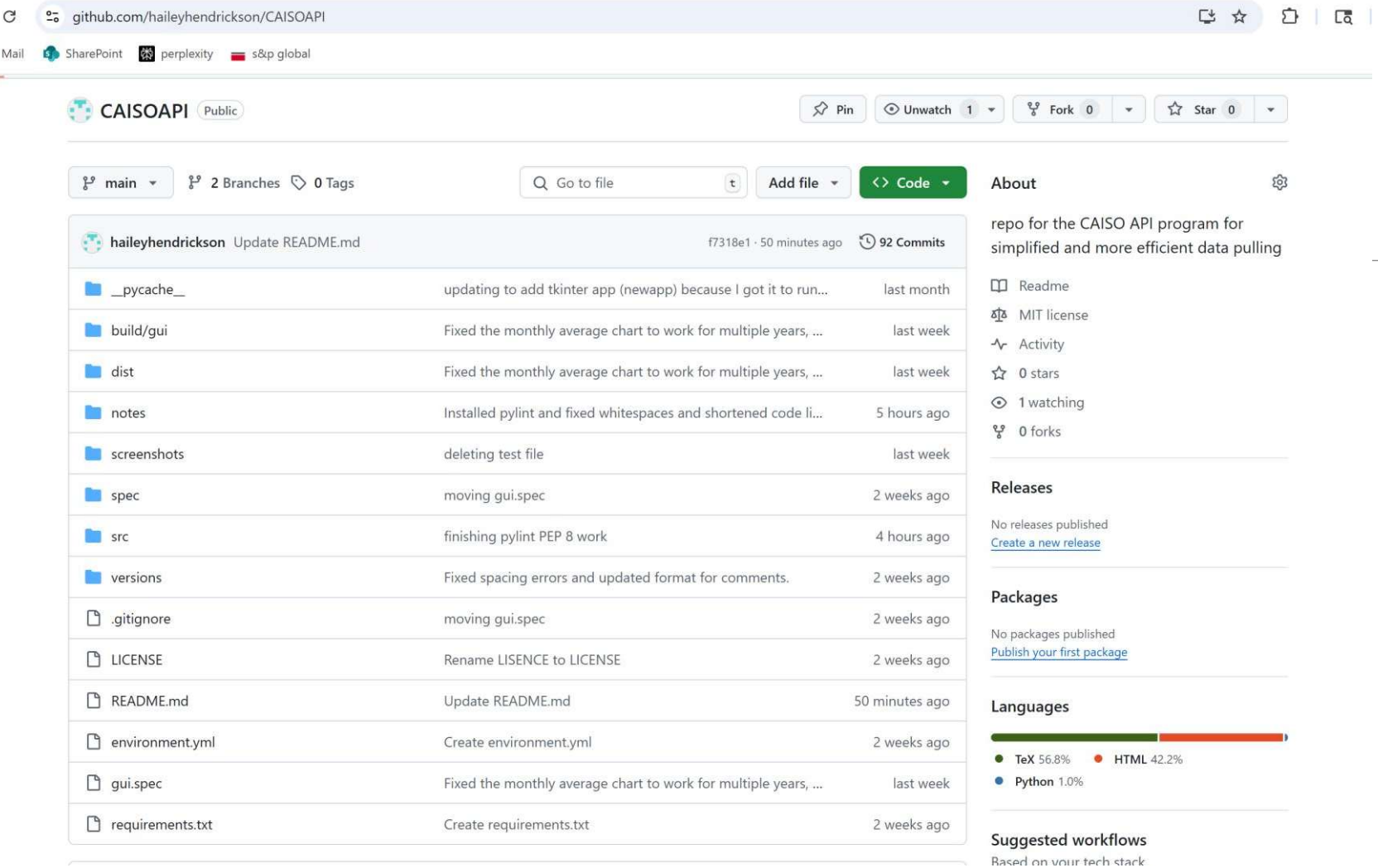
- Separate multiple nodes with a comma.
- All times are set at GMT.
- Each requested report will include the actual report, and the hourly average. (DAM will be the same, as it is already an hourly report).

Last updated: June 2025

Documen

ADDED INSTALLAT
INSTRUCTIONS TO
COMPANY FOLDE

COMMENTED OU
CODE AND UPDAT
WORK LOG.



Documen

CREATED A PROFI
GRADE GITHUB R
INCLUDING:

-INSTALLATION IN

-SOURCE CODE

-DOWNLOADABLE

-SCREENSHOTS O

-LICENSE AND US